

# The Constants of Nature

|                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_{\rm h}:\text{K}=\frac{\text{kelvin}}{\text{K}}\left(\frac{T_{\rm p}}{m_{\rm p}}\right)\left(1-14\,C_{\rm D}^{-4}\,\mathscr{K}_{\rm \Phi}\right)\boxtimes$                            | $m_{\rm de}=\frac{3s}{\mathscr{K}_{\rm S}\,\mathscr{K}_{\rm A}}\left(\frac{16}{4\pi}\right)\left(m_{\rm n}+m_{\rm d}\right)\left(1+\frac{Im(\omega_{\rm i})}{\mathscr{K}_{\rm S}}\left(\frac{L_2}{14}\right)\,\mathscr{K}_{\rm r}^3\right)\boxtimes$                                      | $\mathscr{R}_{\rm K}=\frac{\text{joule}}{\text{K}}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}^2}\right)\left(1-\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_{\rm S}\,\mathscr{K}_{\rm A}\right)\boxtimes$       | $\frac{\mu_{\rm h}}{hc}=\frac{\mathscr{K}_{\rm i}}{4\pi}\left(\frac{t_{\rm p}\,q_{\rm p}}{t_{\rm p}\,m_{\rm e}}\right)\left(1+\sqrt{\frac{1}{D_{\rm D0}}}\,\mathscr{K}_{\rm i}\,\mathscr{K}_{\rm S}\,\mathscr{K}_{\rm A}\right)\boxtimes$ | $c=\left(\frac{L_2}{T_{\rm p}}\right)\left(1+\sqrt{G_{\rm G1}}\left(\mathscr{K}_{\rm i}-\mathscr{K}_{\rm J}\right)\right)\boxtimes$                                                                                                                                                    | $F_{90}=\text{farad}\left(1+\frac{C_{\rm d}}{6}\left(\mathscr{K}_{\rm i}+\mathscr{K}_{\rm J}+\mathscr{K}_{\rm A}\right)\right)\boxtimes$                                                                                                                                                                                                                       | $K_{\rm j}=\frac{\mathscr{K}_{\rm i}}{\pi}\left(\frac{t_{\rm p}\,q_{\rm p}}{t_{\rm p}^2\,m_{\rm p}}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_{\rm i}+\mathscr{K}_{\rm J}+\mathscr{K}_{\rm A}\right)^2\right)\boxtimes$              | $g_{\rm de}=\frac{2\pi}{8\text{K}}\left(1+\frac{16}{t_{\rm p}^2}\left(\mathscr{K}_{\rm i}+\mathscr{K}_{\rm J}+\mathscr{K}_{\rm A}\right)^2\right)\boxtimes$                                                                       |
| $\text{K}:\text{E}_{\rm h}=\frac{\text{kelvin}}{\mathscr{K}_{\rm i}^4}\left(\frac{m_{\rm p}}{t_{\rm p}\,m_{\rm e}}\right)\left(1+14\,C_{\rm D}^{-4}\,\mathscr{K}_{\Phi}\right)\boxtimes$ | $m_{\rm u}=\frac{1}{\mathscr{K}_{\rm S}\,\mathscr{K}_{\rm A}}\sqrt{\frac{16}{4\pi}}\left(m_{\rm n}+m_{\rm d}\right)\left(1+\left(\frac{16}{2\pi}\right)\,\mathscr{K}_{\rm r}^3\right)\boxtimes$                                                                                           | $R_{\rm K}=\frac{2\pi}{\mathscr{K}_{\rm i}^2}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}\,q_{\rm p}^2}\right)\left(1+\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_{\rm S}\,\mathscr{K}_{\rm A}\right)\boxtimes$ | $\frac{\mu_{\rm N}}{hc}=\frac{\mathscr{K}_{\rm i}}{4\pi}\left(\frac{t_{\rm p}\,q_{\rm p}}{t_{\rm p}\,m_{\rm e}}\right)\left(1+\sqrt{\frac{1}{D_{\rm D0}}}\,\mathscr{K}_{\rm i}\,\mathscr{K}_{\rm S}\,\mathscr{K}_{\rm A}\right)\boxtimes$ | $\frac{1}{\text{m}}:\text{Hz}=\frac{1}{\text{meter}}\left(\frac{1}{t_{\rm p}}\right)\left(1+\sqrt{G_{\rm G1}}\left(\mathscr{K}_{\rm i}-\mathscr{K}_{\rm J}\right)\right)\boxtimes$                                                                                                     | $H_{90}=\text{henry}\left(1-\frac{C_{\rm d}}{6}\left(\mathscr{K}_{\rm i}+\mathscr{K}_{\rm J}+\mathscr{K}_{\rm A}\right)\right)\boxtimes$                                                                                                                                                                                                                       | $e=\mathscr{K}_{\rm i}\left(\frac{t_{\rm p}\,q_{\rm p}}{t_{\rm p}^2\,m_{\rm p}}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_{\rm i}+\mathscr{K}_{\rm J}+\mathscr{K}_{\rm A}\right)^2\right)\boxtimes$                                  | $\frac{\mu_{\rm de}}{8\text{K}}=\frac{2\pi}{8\text{K}}\left(1+\frac{16}{t_{\rm p}^2}\left(\mathscr{K}_{\rm i}+\mathscr{K}_{\rm J}+\mathscr{K}_{\rm A}\right)^2\right)\boxtimes$                                                   |
| $\Delta_{\rm vcs}=\frac{E_{\rm i}}{2\pi}\left(\frac{t_{\rm p}}{t_{\rm p}^2\,m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\,\mathscr{K}_{\Phi}^2\right)\boxtimes$                              | $m_{\rm s}=\frac{-2\pi\mathscr{K}_{\rm i}}{\mathscr{K}_{\rm S}}\left(\frac{\text{GeV}}{\sqrt{5}}\right)\left(\frac{t_{\rm p}^2}{t_{\rm p}^2}\right)\left(1+\frac{1}{6^{1/2}}\left(\frac{2\pi}{2s}\right)\,\mathscr{K}_{\rm r}^3\right)\boxtimes$                                          | $Z_0=4\pi\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}\,q_{\rm p}^2}\right)\left(1+\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_{\rm S}\,\mathscr{K}_{\rm A}\right)\boxtimes$                                     | $A_{90}=\text{ampere}\left(1+\sqrt{\frac{9}{J_{0.1}}}\,\mathscr{K}_{\rm i}\,\mathscr{K}_{\rm S}\,\mathscr{K}_{\rm A}\right)\boxtimes$                                                                                                     | $a_0=\frac{1}{\mathscr{K}_{\rm i}^2}\left(\frac{t_{\rm p}\,m_{\rm p}}{m_{\rm e}}\right)\left(1-\sqrt{6\,C_{\rm R1}}\left(\mathscr{K}_{\rm i}-\mathscr{K}_{\rm J}\right)\right)\boxtimes$                                                                                               | $\Omega_{90}=\text{ohm}\left(1-\frac{C_{\rm d}}{6}\left(\mathscr{K}_{\rm i}+\mathscr{K}_{\rm J}+\mathscr{K}_{\rm A}\right)\right)\boxtimes$                                                                                                                                                                                                                    | $\Phi_0=\frac{\pi}{\mathscr{K}_{\rm i}}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}\,q_{\rm p}}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_{\rm i}+\mathscr{K}_{\rm J}+\mathscr{K}_{\rm A}\right)^2\right)\boxtimes$                 | $\frac{\mu_{\rm de}}{\mu_{\rm B}}=\frac{2\pi}{8\text{K}}\left(\frac{m_{\rm p}}{m_{\rm e}}\right)\left(1+\frac{16}{t_{\rm p}^2}\left(\mathscr{K}_{\rm i}+\mathscr{K}_{\rm J}+\mathscr{K}_{\rm A}\right)^2\right)\boxtimes$         |
| $\text{eV}:\text{Hz}=\frac{\text{eV}}{2\pi}\left(\frac{t_{\rm p}}{t_{\rm p}^2\,m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\,\mathscr{K}_{\Phi}^2\right)\boxtimes$                           | $m_{\rm he}=\frac{\sqrt{3}}{\mathscr{K}_{\rm i}^2\,\mathscr{K}_{\rm J}^2}\left(\frac{\text{GeV}}{2\pi}\right)\left(\frac{t_{\rm p}^2}{t_{\rm p}^2}\right)\left(1-\frac{3^{-1}}{\mathscr{K}_{\rm i}\,\mathscr{K}_{\rm S}}\left(\frac{s}{18}\right)\,\mathscr{K}_{\rm r}^3\right)\boxtimes$ | $Z_{\rm p}=\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}\,q_{\rm p}^2}\right)\left(1+\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_{\rm S}\,\mathscr{K}_{\rm A}\right)\boxtimes$                                   | $C_{90}=\text{coulomb}\left(1+\sqrt{\frac{9}{J_{0.1}}}\,\mathscr{K}_{\rm i}\,\mathscr{K}_{\rm S}\,\mathscr{K}_{\rm A}\right)\boxtimes$                                                                                                    | $\sigma_{\rm r}=\sqrt{\frac{32}{18}}\,2\pi\mathscr{K}_{\rm i}^4\left(\frac{t_{\rm p}\,m_{\rm p}}{t_{\rm p}^2\,m_{\rm e}}\right)^2\left(1-2\sqrt{6\,C_{\rm R1}}\left(\mathscr{K}_{\rm i}-\mathscr{K}_{\rm J}\right)\right)\boxtimes$                                                    | $ST_0=\frac{5}{2}+\log\left(\left(\frac{1}{2\pi}\frac{\text{K}\,A_{\rm mass}}{t_{\rm p}^2\,T_{\rm p}\,m_{\rm p}}\right)^{3/2}\frac{\text{K}\,E_{\rm p}}{T_{\rm p}\,P_1}\right)\left(1-\frac{1}{12}\left(\frac{s^2}{\mathscr{K}_{\rm i}\,\mathscr{K}_{\rm J}^2}\right)\left(\mathscr{K}_{\rm i}-\mathscr{K}_{\rm J}-\mathscr{K}_{\rm A}\right)\right)\boxtimes$ | $\frac{h}{e}=\frac{\text{coulomb}}{\mathscr{K}_{\rm i}}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}\,q_{\rm p}}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\mathscr{K}_{\rm i}+\mathscr{K}_{\rm J}+\mathscr{K}_{\rm A}\right)^2\right)\boxtimes$ | $b^{\prime}=\frac{v_{\rm peak}}{2\pi}\left(\frac{1}{t_{\rm p}\,T_{\rm p}}\right)\left(1+\frac{16}{16}\left(\frac{2\pi}{s}\right)^2\left(\mathscr{K}_{\rm i}^2-\mathscr{K}_{\rm J}^2-\mathscr{K}_{\rm A}^2\right)\right)\boxtimes$ |
| $\text{J}:\text{Hz}=\frac{\text{joule}}{2\pi}\left(\frac{t_{\rm p}}{t_{\rm p}^2\,m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\,\mathscr{K}_{\Phi}^2\right)\boxtimes$                         | $\frac{G}{hc}=\text{GeV}^2\left(\frac{t_{\rm p}^2}{t_{\rm p}^2\,m_{\rm p}}\right)^2\left(1-\left(\frac{4\pi}{18}\right)\,\mathscr{K}_{\rm r}^3\right)\boxtimes$                                                                                                                           | $\text{Hz}:\frac{1}{\text{m}}=\frac{1}{\text{second}}\left(\frac{t_{\rm p}}{t_{\rm p}}\right)\left(1+\frac{1}{2}Im\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_{\rm S}\,\mathscr{K}_{\rm A}\right)\boxtimes$                   | $\frac{\mu_{\rm s}^{\prime}}{S}=\frac{2\,\mathscr{K}_{\rm i}\,\mathscr{K}_{\rm J}^2}{S}\left(1-18\left(6\pi\right)\mathscr{K}_{\rm i}\,\mathscr{K}_{\rm S}\,\mathscr{K}_{\rm A}\right)\boxtimes$                                          | $\frac{G_{\rm F}}{(hc)^2}=\sqrt{\frac{32}{18}}\frac{(2\pi)^2}{\text{kilogram}}\left(\frac{\text{joule}}{\text{GeV}}\right)^2\left(\frac{t_{\rm p}^4}{t_{\rm p}^2\,m_{\rm p}}\right)\left(1+28\sqrt{6\,C_{\rm R1}}\left(\mathscr{K}_{\rm i}-\mathscr{K}_{\rm J}\right)\right)\boxtimes$ | $ST_1=\frac{5}{2}+\log\left(\left(\frac{1}{2\pi}\frac{\text{K}\,A_{\rm mass}}{t_{\rm p}^2\,T_{\rm p}\,m_{\rm p}}\right)^{3/2}\frac{\text{K}\,E_{\rm p}}{T_{\rm p}\,P_1}\right)\left(1-\frac{1}{1$                                                                                                                                                              |                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                   |

|                                                                      |                                                                     |
|----------------------------------------------------------------------|---------------------------------------------------------------------|
| $t_p = 5.39125836832313 \times 10^{-40}$ s                           | Planck time                                                         |
| $l_p = 1.61625918175645 \times 10^{-38}$ m                           | Planck length                                                       |
| $q_p = 1.87554596713962 \times 10^{-18}$ C                           | Planck charge                                                       |
| $T_p = 1.41678698590795 \times 10^{32}$ K                            | Planck temperature                                                  |
| $m_p = 2.17642683817579 \times 10^{-8}$ kg                           | Planck mass                                                         |
| $\Xi = 0.999999199973626 \times 10^{-7}$                             | hyperbolic universe boundary                                        |
| 5                                                                    | number of dimensions                                                |
| 7                                                                    | break in scale symmetry                                             |
| $\pi_1 = 0.0854254531533304 \dots$                                   | 1 <sup>st</sup> hyperbolic partition constant                       |
| $\pi_2 = 3.66756753480501 \dots$                                     | 2 <sup>nd</sup> hyperbolic partition constant                       |
| $\pi_3 = 1.87649603900417 \dots + 0.46615262615972 \dots i$          | 3 <sup>rd</sup> hpc                                                 |
| $\pi_4 = 1.87649603900417 \dots - 0.46615262615972 \dots i$          | 4 <sup>th</sup> hpc                                                 |
| $\pi_5 = 4.47826244916751 \dots$                                     | hyperbolic radius constant                                          |
| $\pi_6 = 2.00316562310924 \dots$                                     | hyperbolic radian constant                                          |
| $G_{60} = 1.01494160640965 \dots$                                    | Gieseking's constant                                                |
| $V_6 = 0.29598321281930 \dots$                                       | figure eight knot hyperbolic volume                                 |
| $K = 0.0159655541772190 \dots$                                       | Catalan's constant                                                  |
| $C_d = 0.2916506090430818 \dots$                                     | domino tiling constant                                              |
| $C_4 = 1.79162281206959 \dots$                                       | dimer constant                                                      |
| $\omega_1 = 0.764974918528596 \dots + 1.3249706214087 \dots i$       | omega_1 constant                                                    |
| $\omega_2 = 1.5295940370597192 \dots$                                | omega_2 constant                                                    |
| $i^{i^{i^i}}$                                                        | power tower of i                                                    |
| $i = (-1)^{1/2}$                                                     | imaginary unit                                                      |
| $\phi_i = (-1)^{1/2}$                                                | imaginary golden ratio                                              |
| $G_9 = 1.51872847301812 \dots$                                       | tether length for grazing half the unit circle                      |
| $\pi = 3.14159265358979 \dots$                                       | Archimedes' constant                                                |
| $S = 2.29558714939263 \dots$                                         | universal parabolic constant                                        |
| $s = 5.244115108584261 \dots$                                        | arc length of the unit lemniscate                                   |
| $L = 2.622057554429119 \dots$                                        | lemniscate constant                                                 |
| $L_1 = 1.31102877146205 \dots$                                       | 1 <sup>st</sup> lemniscate constant                                 |
| $L_2 = 0.599070117367796 \dots$                                      | 2 <sup>nd</sup> lemniscate constant                                 |
| $C_{ij} = 0.884213084793979 \dots$                                   | ubiquitous constant                                                 |
| $G_{66} = 0.834626841674073 \dots$                                   | Gauss's constant                                                    |
| $C_{CF} = 0.697774657964819 \dots$                                   | continued fraction constant                                         |
| $R_{FR} = 0.9981360445980509 \dots$                                  | Ramanujan's 1 <sup>st</sup> continued fraction constant             |
| $F = 1.2267421047027035 \dots$                                       | Fibonacci factorial constant                                        |
| $M = -1.477456459463318 \dots$                                       | Modeling constant                                                   |
| $e = 2.71828182845904 \dots$                                         | Euler's number                                                      |
| $\gamma = 0.577215664901532 \dots$                                   | Euler-Mascheroni constant                                           |
| $S = 0.82282549678847 \dots$                                         | Sierpinski's constant                                               |
| $C_{JP} = 2.4048255610595772 \dots$                                  | 1 <sup>st</sup> root of the Bessel function                         |
| $f_{J11} = 1.19967864025773 \dots$                                   | Real fixed point of the hyperbolic cotangent                        |
| $D_{N0} = 0.739085133216160 \dots$                                   | Dottie number                                                       |
| $L_1 = 0.662743419341981 \dots$                                      | Laplace limit                                                       |
| $a_F = 2.50290787590589 \dots$                                       | alpha Feigenbaum constant                                           |
| $\delta = 4.66920160910299 \dots$                                    | delta Feigenbaum constant                                           |
| $S = 3.246479603371714 \dots$                                        | silver constant                                                     |
| $p = 1.3247179574474 \dots$                                          | plastic constant                                                    |
| $C_Q = 0.605443657196732 \dots$                                      | QRS constant                                                        |
| $q_{QA} = 2.03816937970215 \dots$                                    | associated QRS constant                                             |
| $\kappa_{\text{HSM}} = 0.3532363781584995 \dots$                     | Hafner-Sarnak-McCurley constant                                     |
| $T_0 = 2.731500000000000 \dots \times 10^5$ kelvin                   | freezing point of water                                             |
| $p = 1.000000000000000 \dots \times 10^2$ pascal                     | 1 bar                                                               |
| $P_1 = 1.01325003754773 \dots \times 10^5$ pascal                    | standard atmospheric pressure                                       |
| $E_1 = 6.09110229711386 \dots \times 10^{-24}$ joules                | energy associated with $\Delta v_{\text{CS}}$                       |
| $f_1 = 5.400000000000000 \dots \times 10^{14}$ Hz                    | frequency of max light wavelength                                   |
| $\lambda_{\text{peak}} = 4.96511423174427 \dots$                     | peak $\lambda$ of spectral radiance/unit wavelength                 |
| $\nu_{\text{peak}} = 3.24149397212207 \dots$                         | peak emission of spectral flux/unit frequency                       |
| $\log(x)$ = natural logarithm                                        |                                                                     |
| $\Gamma(x)$ = gamma function                                         | ln = lumen = cd · sr                                                |
| $\zeta(x)$ = Riemann zeta function                                   | $N$ = newton = m kg/s <sup>2</sup>                                  |
| $\ln$ = derangement function                                         | $N$ = noether = m kg/s                                              |
|                                                                      | $\rho$ = ohm = m <sup>2</sup> kg/s <sup>2</sup> C <sup>2</sup>      |
| <b>units</b>                                                         | $\text{Pa} = \text{pascal} = \text{kg/m}^2 \text{ s}^2$             |
| $m$ = meter                                                          | $S$ = siemens = s C <sup>2</sup> /m <sup>2</sup> kg                 |
| $m$ = second                                                         | $T$ = tesla = kg/s <sup>2</sup> C                                   |
| $C$ = coulomb                                                        | $V$ = volt = m <sup>2</sup> kg/s <sup>2</sup> C                     |
| $K$ = kelvin                                                         | $W$ = watt = m <sup>2</sup> kg/s <sup>3</sup>                       |
| $\text{kg}$ = kilogram                                               | $\text{Wh} = \text{weber} = \text{m}^2 \text{ kg/s} \cdot \text{C}$ |
|                                                                      |                                                                     |
| <b>derived units</b>                                                 |                                                                     |
| MHz = megahertz = 1.000000000000000 ... × 10 <sup>6</sup> 1/s        |                                                                     |
| fm = femtometer = 1.000000000000000 ... × 10 <sup>-15</sup> m        |                                                                     |
| eV = electron-volt = 1.000000000000000 ... eV                        |                                                                     |
| MeV = megaelectron-volt = 1.000000000000000 ... × 10 <sup>6</sup> eV |                                                                     |
| GeV = gigaelectron-volt = 1.000000000000000 ... × 10 <sup>9</sup> eV |                                                                     |
| $r_e = \frac{4\pi}{(r_s)^2}$                                         | $r_n = \frac{r_n}{r_s}$                                             |
| $r_s = \frac{r_s}{r_n}$                                              | $r_n = \frac{r_n}{r_s}$                                             |
| $r_e = \frac{r_e}{r_n}$                                              | $r_n = \frac{r_n}{r_s}$                                             |
| $r_s = \frac{r_s}{r_n}$                                              | $r_n = \frac{r_n}{r_s}$                                             |
| $r_e = \frac{r_e}{r_n}$                                              | $r_n = \frac{r_n}{r_s}$                                             |
| $r_s = \frac{r_s}{r_n}$                                              | $r_n = \frac{r_n}{r_s}$                                             |
| $r_e = \frac{r_e}{r_n}$                                              | $r_n = \frac{r_n}{r_s}$                                             |
| $r_s = \frac{r_s}{r_n}$                                              | $r_n = \frac{r_n}{r_s}$                                             |
| $r_e = \frac{r_e}{r_n}$                                              | $r_n = \frac{r_n}{r_s$                                              |